Enzymes

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Enzymes [1]

Which statement describes a catalyst?

A. a substance that is used up as it speeds up a chemical reaction

B. a substance that increases the rate of a chemical reaction and is not changed by the reaction

C. a substance that slows down a chemical reaction and is not changed by the reaction

D. a substance that increases the rate of a chemical reaction by becoming part of the product

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2 · Part 1 Core Enzymes [1]

Which statement about enzymes is correct?

A. They are carbohydrates that act as biological catalysts.

B. They are proteins that are used up in metabolic reactions.

C. They are lipids that speed up digestive reactions only.

D. They are proteins that act as biological catalysts in metabolic reactions.

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3 · Part 1 Core Enzymes [1]

Why are enzymes needed in all living organisms?

A. Without enzymes, metabolic reactions would be too slow to sustain life.

B. Enzymes provide the energy that metabolic reactions use up.

C. Enzymes are the only source of protein in cells.

D. Enzymes raise body temperature so that reactions go faster.

4 · Part 1 Core Enzymes [1]

Which statement about how an enzyme works is correct?

A The product has a shape that is complementary to the substrate.

B The enzyme is used up when the products are formed.

C The substrate fits into the active site because their shapes are complementary, and then products are formed.

D The active site changes shape permanently each time a product is formed.

5 · Part 1 Core Enzymes [1]

The rate of an enzyme-catalysed reaction was measured at different temperatures. All other conditions were kept the same.

temperature / °C	rate of reaction / arbitrary units
10	8
20	18
30	34
40	52
50	30
60	2

Which conclusion is supported by these results?

A Of the temperatures tested, the enzyme was most active at 40 °C.

B The enzyme was denatured at 30 °C.

C The rate of reaction increased every time the temperature was raised.

D The enzyme had its highest activity at 60 °C.

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6 · Part 1 Extended Enzymes [1]

An enzyme from the stomach has an optimum pH of 2. The enzyme is mixed with its substrate in a solution at pH 8, at its optimum temperature. No product is formed.

Which explanation is correct?

A At pH 8, the enzyme and substrate molecules have less kinetic energy, so they collide less often.

B At pH 8, the enzyme is used up as it reacts with the substrate.

C At pH 8, the substrate changes shape so that it fits the active site more tightly.

D At pH 8, the shape of the active site has changed, so the substrate no longer fits.

7 · Part 1 Extended Enzymes [1]

Which sequence describes enzyme action?

A enzyme + product → enzyme-product complex → enzyme + substrate

B enzyme + substrate → enzyme-substrate complex → enzyme + products

C enzyme + substrate → enzyme-substrate complex → denatured enzyme + products

D substrate + product → enzyme-substrate complex → enzyme

8 · Part 1 Extended Enzymes [1]

Lipase breaks down fat molecules. It does not break down starch molecules.

Which statement explains this?

A Starch molecules do not collide with lipase molecules.

B Lipase is denatured by starch.

C Fat molecules, but not starch molecules, have a shape complementary to the active site of lipase.

D Lipase is used up after breaking down fat, so none is left to break down starch.

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9 · Part 1 Extended Enzymes [1]

The optimum temperature of an enzyme is 40 °C. The temperature of a reaction catalysed by this enzyme is increased from 20 °C to 35 °C. The rate of reaction increases.

Why does the rate of reaction increase?

A The enzyme and substrate molecules gain kinetic energy, so there are more frequent effective collisions.

B The active site changes shape so that it becomes more complementary to the substrate.

C Some enzyme molecules are denatured, so more active sites are exposed.

D More enzyme molecules are made as the temperature rises.

10 · Part 1 Extended Enzymes [1]

The optimum temperature of an enzyme is 37 °C. Two samples of the enzyme were treated in different ways.

Sample 1 was kept at 70 °C for 10 minutes and then cooled to 37 °C.

Sample 2 was kept at 5 °C for 10 minutes and then warmed to 37 °C.

Substrate was then added to each sample at 37 °C. Sample 1 formed no product. Sample 2 formed product at the normal rate.

Which explanation is correct?

A At 5 °C, sample 2 was denatured, but its active site changed back when it was warmed.

B At 70 °C, sample 1 had too little kinetic energy to form enzyme-substrate complexes.

C At 70 °C, the active site of sample 1 changed shape permanently, but sample 2 was not denatured at 5 °C.

D At 70 °C, sample 1 was used up by reacting with the water in the solution.

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11 · Part 1 Core Enzymes [5]

Each cell in the human liver contains thousands of different enzymes. Enzymes are described as biological catalysts.

(a) State what is meant by the term catalyst. [2]

(b) State the type of biological molecule that enzymes are made of. [1]

(c) Explain why all living organisms need enzymes. [2]

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12 · Part 1 Core Enzymes [5]

Milk contains a sugar called lactose. In the small intestine, the enzyme lactase breaks down each lactose molecule into one glucose molecule and one galactose molecule.

(a) Name the substrate and the products of this reaction. [2]

(b) Describe how lactase breaks down lactose. Refer to the active site in your answer. [3]

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13 · Part 1 Core Enzymes [7]

In an investigation, a student mixed amylase solution with starch solution in five test-tubes. Each test-tube was kept in a water bath at a different temperature.

Every 30 seconds the student added one drop of each mixture to iodine solution. Iodine solution stays orange-brown when no starch is present. The student recorded the time taken for all the starch to be broken down.

temperature / °C	time taken for all the starch to be broken down / s
10	600
25	240
40	90
55	300
70	starch still present after 900

(a) State the temperature at which the amylase was most active in this investigation. [1]

(b) Describe the effect of temperature on the activity of amylase shown by these results. [3]

(c) Explain the result at 70 °C. [2]

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(d) State one variable, other than temperature, that the student should keep the same in every test-tube. [1]

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14 · Part 1 Core Enzymes [5]

Cooked egg white is a protein. When it is mixed with water it forms a cloudy suspension. The suspension becomes clear when a protease enzyme breaks down the protein.

In an investigation, a student added the same volume of protease solution to the same volume of egg-white suspension at six different pH values. All tubes were kept at 37 °C. The student recorded the time taken for each suspension to become clear.

pH	time taken to become clear / min
1	12
2	6
3	8
5	20
7	more than 30
9	more than 30

(a) State the optimum pH of this protease from these results. [1]

(b) State how the results show that the reaction at pH 3 was slower than the reaction at pH 2. [1]

(c) Explain why the suspension at pH 9 did not become clear. [2]

(d) The student repeated the whole investigation two more times. Suggest why. [1]

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15 · Part 1 Core Enzymes [6]

Some fish live in seas near Antarctica, where the water temperature stays close to 0 °C. The body temperature of these fish is the same as the temperature of the water.

Scientists measured the activity of the same type of enzyme taken from one of these fish and from humans.

temperature / °C	activity of fish enzyme / arbitrary units	activity of human enzyme / arbitrary units
0	60	5
10	85	15
20	50	30
30	10	55
40	0	90
50	0	20

(a) State the optimum temperature of each enzyme shown by these data. [2]

(b) Compare the activity of the two enzymes between 0 °C and 20 °C. [2]

(c) Suggest why it is important for these fish that their enzyme has high activity at low temperatures. [2]

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16 · Part 1 Extended Enzymes [6]

Maltose is a sugar made of two glucose units joined together. The enzyme maltase breaks down maltose into glucose.

(a) Explain how maltase catalyses the breakdown of maltose. Use the terms active site, enzyme-substrate complex and product in your answer. [4]

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(b) Sucrose is another sugar made of two sugar units joined together. Maltase does not break down sucrose.

Explain why maltase does not break down sucrose. [2]

17 · Part 1 Extended Enzymes [7]

Potato tissue contains an enzyme that breaks down hydrogen peroxide into water and oxygen.

In an investigation, a student added potato extract to hydrogen peroxide solution at six different temperatures. The student collected the oxygen released in 2 minutes at each temperature.

temperature / °C	volume of oxygen collected in 2 minutes / cm^3
15	6.0
25	12.0
35	21.0
45	27.0
55	9.0
65	0.0

(a) Calculate the percentage increase in the volume of oxygen collected when the temperature was increased from 15 °C to 35 °C. [2]

(b) Explain, in terms of kinetic energy and collisions, why the volume of oxygen collected increased between 15 °C and 45 °C. [3]

(c) Describe how the student could change the investigation to find the optimum temperature of the enzyme more precisely. [2]

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18 · Part 1 Extended Enzymes [5]

Enzyme X and enzyme Y are two digestive enzymes. Their activity was measured at different pH values. All other conditions were kept the same.

pH	activity of enzyme X / % of maximum	activity of enzyme Y / % of maximum
2	95	0
4	40	10
6	5	60
8	0	100
10	0	45

(a) Identify the pH values in the table at which both enzymes were active. [1]

(b) Explain, in terms of shape and fit, why enzyme X has no activity at pH 8. [3]

(c) Enzyme Y was kept at pH 2 for 30 minutes. It was then put into a solution at pH 8 with its substrate. Its activity was still 0%.

Suggest what this result shows about the effect of pH 2 on enzyme Y. [1]

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19 · Part 1 Extended Enzymes [6]

In an investigation, a student mixed each of three enzymes, P, Q and R, with each of three substrates. All mixtures were kept at the optimum temperature and pH of the enzymes. The results are shown.

enzyme	starch	protein	fat
P	broken down	not broken down	not broken down
Q	not broken down	broken down	not broken down
R	not broken down	not broken down	broken down

(a) State the property of enzymes that these results show. [1]

(b) In a different organism, a change in the gene for enzyme P produces enzyme P with a different shape of active site.

Suggest and explain the effect of this change on the breakdown of starch. [2]

(c) The student thinks that enzyme P is used up when it breaks down starch.

Describe how the student could use the mixture of enzyme P and starch to show that enzyme P is not used up. [3]

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20 · Part 1 Extended Enzymes [7]

An enzyme was mixed with a large amount of its substrate at 30 °C and at 60 °C. The mass of product formed was measured every minute for 5 minutes. At the end of the investigation, a lot of substrate was still present in both mixtures.

time / min	mass of product at 30 °C / mg	mass of product at 60 °C / mg
0	0	0
1	4	9
2	8	13
3	12	14
4	16	14
5	20	14

(a) Calculate the mean rate of product formation at 30 °C over the 5 minutes. Give the unit. [2]

(b) Compare the formation of product at the two temperatures. [2]

(c) Explain why no more product was formed after 3 minutes at 60 °C. [3]

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